

AI RESUME ANALYZER

Submitted by: Omkar Pujar

Course: B.E. Robotics and Automation

University: Visvesvaraya Technological University

USN: 2VX22RA011

ABSTRACT:

The AI Resume Analyzer is a web-based application developed to automate the process of resume evaluation. In today's competitive job market, candidates often struggle to assess the quality of their resumes and understand the skills required by recruiters. This system helps bridge that gap by providing instant analysis and feedback.

The application allows users to upload resumes in PDF format. The system extracts textual content from the document and processes it using rule-based Natural Language Processing techniques. It identifies key skills, evaluates missing skills, calculates a resume score, and provides improvement suggestions.

Additionally, the system suggests suitable job roles based on the detected skills. The project is implemented using Python and Streamlit, providing an interactive and user-friendly interface.

INTRODUCTION:

In modern recruitment processes, resumes serve as the first point of interaction between candidates and recruiters. A well-structured resume significantly increases the chances of getting shortlisted. However, many candidates lack the knowledge required to create effective resumes.

Manual resume evaluation is time-consuming and inconsistent. Recruiters often rely on automated systems to filter candidates based on specific criteria. Therefore, it is important for candidates to understand how their resumes are evaluated.

The AI Resume Analyzer is designed to automate resume analysis and provide structured feedback. It helps users identify their strengths and weaknesses and guides them in improving their resumes according to industry standards.

OBJECTIVES:

- To develop an automated resume analysis system
- To extract text from PDF resumes
- To identify technical and soft skills
- To detect missing skills
- To calculate a resume score
- To suggest improvements
- To recommend suitable job roles
- To create a user-friendly interface

PROBLEM STATEMENT:

Many job seekers face difficulties in preparing effective resumes due to lack of guidance and feedback.

Common problems include poor formatting, missing skills, and lack of clarity about industry requirements.

Existing solutions are often expensive, complex, or not accessible to beginners. Hence, there is a need for a simple and intelligent system that provides automated resume evaluation and feedback.

SCOPE OF THE PROJECT:

The system focuses on:

- Resume upload in PDF format
- Text extraction
- Skill identification
- Resume scoring
- Suggestions generation
- Job role recommendation

Limitations:

- Works best with structured resumes
- Uses rule-based logic

TECHNOLOGY STACK:

Programming Language: Python

Framework: Streamlit

Libraries Used:

- PyPDF2
- Streamlit

METHODOLOGY:

Steps involved:

1. Upload resume
2. Extract text
3. Process text
4. Detect skills
5. Identify missing skills
6. Calculate score
7. Suggest job role
8. Display result

Flow Chart:

Start → Upload → Extract → Analyze → Score →
Output → End

IMPLEMENTATION:

The project consists of three main files:

- app.py (User Interface)
- agent.py (Analysis Logic)
- utils.py (PDF Extraction)

FEATURES:

- Resume upload
- Skill detection
- Missing skill identification
- Resume scoring
- Job role suggestion
- Download report

ANALYSIS LOGIC:

The system uses keyword-based matching to detect skills.

Skill Detection:

- Python
- Java
- React
- Robotics

Score Calculation:

Score = 50 + (number of skills × 5)

Job Role Suggestion:

- React → Frontend Developer
- Python → Software Developer
- Robotics → Robotics Engineer

RESULTS:

The system provides:

- Resume score
- Key skills
- Missing skills
- Suggestions
- Job role

Image:



